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Attention Deficit Hyperactivity Disorder (ADHD) and Adolescent Development

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Office of Academic Affairs

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Attention Deficit Hyperactivity Disorder (ADHD)

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Session Objectives

1. Recognize the core symptoms and various presentations of ADHD.
2. Define the diagnostic criteria for ADHD.
3. Outline the steps to evaluating a child for ADHD.
4. Compare and contrast the diagnostic criteria for the 4 types of ADHD.
5. Describe the comorbidities that are common in children with ADHD and recognize the need to evaluate for these.
6. Summarize the components of ADHD treatment and when each is indicated based on patient age.

Source: Course Syllabus

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Epidemiology of ADHD

- Estimated to affect almost 9% of children
- **Boys are 2X more likely to be diagnosed with ADHD than Girls**
- There is no association with a specific race or ethnicity
- **The incidence of ADHD is increased in children with neurologic, developmental, psychiatric, and genetic disorders** (e.g., Tuberous Sclerosis, Autism Spectrum Disorder, Anxiety, Fragile X Syndrome)

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Risk Factors for ADHD

- ADHD risk factors involve an interplay between genetic, biological, and environmental factors
- Environmental Factors: **Prenatal Cigarette Smoke Exposure**, Alcohol, **Lead Exposure**
- Biological Factors: **Low Birth Weight** and Prematurity

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Presentation of ADHD

- ADHD is a heterogenous neurodevelopmental disorder characterized by symptoms of **inattention, motor hyperactivity, and impulsivity** that is **inappropriate for developmental stage**
- ADHD presents with a different ratio of inattention and hyperactivity
- Symptoms become more manageable with age; however, up to 60% of adults continue to have symptoms that interfere with daily living

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Criteria for ADHD Diagnosis

- Persistent pattern of inattentiveness and/or hyperactivity-impulsivity that interferes with functioning or development
    - **Six or more symptoms** from either (or both if combined) category must be present for **at least 6 months** that is inconsistent with the developmental level and **causes functional impairment**
- Symptom onset must be **prior to 12 years of age**
- Symptoms must be present in **two or more settings** (e.g., home, school, work, etc.)
- Symptoms cannot be explained by another disorder

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The Four Types of ADHD

1. **ADHD Inattention**: 6 out of 9 Inattentive Behaviors persists for at least 6 months
2. **ADHD Hyperactive/Impulsive**: 6 out of 9 Hyperactive/Impulsive Behaviors persist for at least 6 months
3. **ADHD Combined**: 6 out of 9 Inattentive Behaviors **PLUS** 6 out of 9 Hyperactive/Impulsive Behaviors persist for at least 6 months
4. **ADHD Other Specified or Unspecified**: Symptoms are clinically significant but do not meet the "6 out of 9" criteria for any category

Think "6 out of 9" for "Six Months"

Inattentive Behaviors	Hyperactive and Impulsive Behaviors
a. Fails to give close attention to details or makes careless mistakes during activities	a. Fidgets with or taps hands or feet; squirms in seat
b. Has difficulty sustaining attention in tasks or play activities	b. Leaves seat in situations when remaining seated is expected (e.g., leaves chair in classroom)
c. Does not seem to listen when spoken to directly	c. Runs about or climbs in situations where it is inappropriate (NOTE: in adolescents or adults this may be described as "feeling restless")
d. Does not follow through on instructions and fails to finish chores or schoolwork	d. Unable to play or engage in leisure activities quietly
e. Has difficulty organizing tasks and activities	e. Often "on the go" as if "driven by a motor"
f. Avoids, dislikes, or is reluctant to engage in tasks that require sustained mental effort (e.g., homework)	f. Talks excessively
g. Loses thigs necessary for tasks or activities (e.g., pencils, tools, wallets, cellphone)	g. Blurts out answer before a question has been completed
h. Easily distracted by extraneous stimuli (e.g., unrelated thoughts)	h. Has difficulty waiting for his or her turn
i. Forgetful in daily activities (e.g., keeping appointments, doing chores)	i. Interrupts or Intrudes on others (e.g., uses other people's things without asking for permission)

Steps to Evaluating A Child for ADHD

Patient History

Physical Examination

ADHD Rating Scale

Steps for Evaluating ADHD: Patient History

- Diagnosis relies on a comprehensive evaluation focused on patient history—there is no single diagnostic test
  - The perspective of both **home behavior AND school behavior must be accounted for**
- Be sure to obtain family history, social history, and school history

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Steps for Evaluating ADHD: Physical Examination

- Vital Signs: It is important to establish baseline height, **weight, BP, and pulse** (ie. stimulants can increase HR, increase BP, and decrease appetite to cause weight loss)
- During the physical exam be sure to assess language development, behavior, and social interaction
- **<u>ALWAYS assess for and rule out hearing and vision deficits</u>**-–often these can present as problems with sustaining attention or impulsivity especially in the school setting

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Steps for Evaluating ADHD: Rating Scales

- Often ADHD symptoms may not be present during the office visit—therefore rating scales are very important
    - Helps determine baseline behavior across various settings (e.g., Vanderbilt Rating Scale has Teacher and Parent Forms)
- Rating Scales are based on the DSM-V Criteria and help to assess the degree of functional impairment
- No rating scale should be used in isolation to make a diagnosis (ie. **Rating scales are NOT diagnostic**)
- Rating scales only document the presence of symptoms of inattention, hyperactivity, and impulsivity—they do not necessarily determine if these symptoms are attributable to ADHD

Comorbidities

- Emotional or Behavioral Comorbidities: **Oppositional Defiant Disorder**, **Conduct Disorder**, Depression, Anxiety, Substance Use, Intermittent Explosive Disorder
- Developmental Comorbidities: Specific Learning Disorders, Language Disorders, Autism Spectrum Disorder
- Other Comorbidities: Sleep Disorders, Tic Disorders

It is important to have a low threshold when assessing for comorbidities—not every behavior will be caused by ADHD

Management of ADHD

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- Treatment for ADHD involves a combination of behavioral interventions, educational interventions (IEPs), and medication
- Treatment is guided by the patient's age:
  - **Children Under 6: Parent Training Behavioral Management (PTBM) is first line**
    - If medication is necessary → always give **methylphenidate** rather than amphetamines
  - **Children Over 6: Medication is first line**; Supplemented with Behavioral Modifications
- You may refer for a speech/language evaluation, psychologist, or developmental behavioral pediatrician for further management

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Adolescent Development and Adolescent Care

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Session Objectives

1. Discuss and recognize the key features and subtypes of adolescent development.
2. Explain the focus of the adolescent well visit: specifically, regarding a thorough psychosocial history, anticipatory guidance, and screening for risk-taking behaviors.
3. Understand the areas of adolescent care that can and cannot remain confidential.

Source: Course Syllabus

Adolescence Defined

- A critical period that is characterized as **gradual development towards autonomy and adult decision making**
- Between the ages of 11 – 21 years of age
  - Early Adolescence: Ages 11 – 14
  - Middle Adolescence: Ages 15 – 17
  - Late Adolescence: Ages 18 – 21
- Adolescents are a unique and vulnerable patient population
- Unmet emotional and medical needs during this transition towards adulthood predict poor health outcomes and a lower quality of life in adulthood

Development in Adolescence

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- The four key areas of development that occur in adolescence include **biological** development, **cognitive** development, **psychosocial** development, **emotional** development

- Cognitive Development: Concrete Thinking → Abstract Thinking → Formal Operational Thinking
- Psychosocial Development: Develop Autonomy, Establish Identity, Develop Future Orientation
- Emotional Development: Emotional and Social Competence

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Asynchrony in Development

- It is important to understand that emotional development often develops after the other areas of development
  - Example: emotional-physical asynchrony (ie. physical development occurs at a quicker rate than emotional) which causes adolescents to be treated older than their emotional stage of development
- This **asynchronous development can lead to increased risk-taking behaviors** (areas of the brain responsible for impulse control are still developing)
  - Risk Taking Behaviors: Alcohol, Drugs, Tobacco/Vaping, Unprotected Sex

It is very important to perform preventative counseling and establish trust with the adolescent patient

Mental Health in Adolescence

- 20% of adolescents have a mental health disorder with more than 50% of lifetime cases of mental illness beginning at the age of 14
- Patients aged 10-24 years old account for 14% of all suicides
- **Suicide is the 2nd leading cause of death for patients aged 10-14 years old**
    - Patients who identify as sexual minorities have even higher rates of suicide and suicide attempts
- Youth and young adults account for the highest percentage of Emergency Department visits relating to self harm

It is critically important to understand the emotional and physical turbulence adolescent patients go through in order to provide a safe and trusted medical home to our patients

Caring for an Adolescent Patient

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- Performing a thorough psychosocial history and screening for risk-taking behaviors are two primary goals for the adolescent well-visit
  - **<u>HEEADSSS</u>**: **H**ome, **E**ducation/**E**mployment, **E**ating/**E**xercise, **A**ctivities, **D**rugs, **S**exuality, **S**uicide (Depression), **S**afety
- Whenever possible, **try to talk to adolescent patients alone**-–this allows you to establish trust with your patient while encouraging their sense of autonomy and independence
- If your adolescent patient admits to risk taking behavior, **thank them for sharing and provide anticipatory guidance** based on further questioning

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Confidentiality in Adolescence

- Consent and Confidentiality for adolescents vary by state and specific service
  - Physicians should deliver confidential health services in situations involving **reproductive health, sexuality, gender identity and expression, substance use, and mental health** to consenting adolescents
- Emancipated Minors can consent for themselves for all treatments

- Confidentiality EXCEPTIONS (these must all be reported to the proper authorities):
  - information suggesting **someone is in imminent danger**
  - suspicion **of sexual or physical abuse** of a minor
  - the diagnosis of certain **communicable diseases**

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Summary

- ADHD is a neurodevelopmental disorder which causes functional impairment and requires specific criteria for diagnosis:
    - At least 6 out of 9 symptoms present for at least six month
    - Symptoms need to be present before 12 years of age
    - Symptoms need to be present in two or more settings (e.g., school and home)
    - Cannot be explained by another disorder
- Adolescence is a unique and vulnerable transitional period for many patients
    - The primary goals for the adolescent well visits include anticipatory guidance, taking a thorough psychosocial history, and screening for risk-taking behaviors
- Confidentiality should be expected for consenting adolescents in matters of reproductive health, sexuality, gender identity, substance use, and mental health

Question #1:

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A 9-year-old male is brought into your office by his mother who states that the child has not been doing well in school for the past year. He often speaks out of turn and frequently stands up while doing his schoolwork. The patient has been sent to the principal's office on multiple occasions for talking to his classmates during lessons. His grades are poor in part because the child often comes unprepared for class. The patient often apologizes for forgetting his materials and appears frustrated when he forgets to bring in his homework. At home, the patient often refuses to engage in homework and is often found playing video games when he is supposed to be completing his chores. Which of the following is the most likely diagnosis?

A. Anxiety Disorder
B. Autism Spectrum Disorder
C. Attention Deficit Hyperactivity Disorder
D. Oppositional Defiant Disorder
E. Conduct Disorder

Question #1 Answer:

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D. Oppositional Defiant Disorder
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Question #2:

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A 40-year-old male presents to you for an initial visit at your clinic. He has no physical complaints. The patient currently works as a waiter. As a child, the patient had trouble in school causing him to repeat first and second grade. His teachers often complained to his father that he was unable to sit still in class and was disruptive. As an adult, the patient has bounced from job to job. His most recent job was in construction; however, he recently lost this job due to difficulty keeping track of the building materials. The patient admits he gets bored easily and admits that he often cannot help saying what is on his mind as soon as an idea enters his head. His family history is positive for a specific learning disorder in his brother. Physical examination is within normal limits. What is the most likely diagnosis?

A. Antisocial Personality Disorder
B. Specific Learning Disorder
C. Major Depressive Disorder
D. Attention Deficit Hyperactivity Disorder
E. Intellectual Disability

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Question #3:

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An 8-year-old boy is brought to his pediatrician by his parents due to concerns about his recent behavior. They complain that he often avoids doing chores around the house and he must be told to repeatedly do his homework. When told firmly to complete these tasks, he often objects, insists he would prefer to play his video games, but eventually is able to complete his homework without any difficulty. His school performance is slightly below average, and his teacher mentioned that he may not be reaching his full potential. His past medical history is insignificant. What is his most likely diagnosis?

A. Attention Deficit Hyperactivity Disorder
B. Specific Learning Disorder
C. Normal Age-Appropriate Behavior
D. Oppositional Defiant Disorder
E. Anxiety Disorder

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Question #5:

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A 15-year-old female presents to the office for her annual well visit. She has no specific concerns and reports that she is getting good grades. When asked about her sexual history, she admits that she has had a boyfriend for three months. She does not want to tell her parents about the relationship because she knows they do not want her to date. She recently began to have sexual intercourse with this male partner and would like to start an oral contraceptive. The patient has no other medical issues and is not currently on any medications. Which of the following is the most appropriate initial response for you to give?

A. "I can only prescribe you oral contraceptive pills with your parent's consent"
B. "I can prescribe you oral contraceptives, but I will also have to tell your parents you are sexually active"
C. "You are too young to be having sexual intercourse and I recommend you stop"
D. "I can give you a prescription for oral contraceptives, but I also recommend that you talk about your relationship with your parents when you feel comfortable to do so"
E. "You should really tell your parents that you are sexually active"

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